EDITORIAL

SCIENCE BEHIND THE STUDY

Advances in Preventing Transmission of Mitochondrial DNA Diseases

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Hyslop et al.[1] report in the *Journal* their experience with mitochondrial donation as a strategy to reduce the risk of transmitting mitochondrial DNA (mtDNA) diseases (see Key Concepts). In a related article, McFarland et al.[2] describe the associated reproductive care pathway, established in Newcastle, United Kingdom, after the U.K. government legalized mitochondrial donation in 2015.

DISORDERS CAUSED BY MTDNA PATHOGENIC VARIANTS

Pathogenic, damaging variants in mtDNA are responsible for severe maternally inherited disorders with a high risk of transmission, including MELAS (mitochondrial encephalopathy with lactic acidosis and strokelike episodes) syndrome, mainly caused by the m.3243A→G variant in *MT-TL1*; Leber's hereditary optic neuropathy, mainly caused by the m.11778G→A variant in *MT-ND4*; and NARP (neuropathy, ataxia, and retinitis pigmentosa) syndrome, resulting from an m.8993T→G or m.8993T→C variant in *MT-ATP6*. The clinical phenotypes associated with these variants are highly heterogeneous, largely owing to variations in heteroplasmy loads (the proportion of mtDNA that is variant) within mitochondria, cells, and tissues. The onset and severity of symptoms in mtDNA diseases depend on the level of heteroplasmy, with clinical manifestations appearing once a variant-specific threshold is exceeded.[3] Higher levels of heteroplasmy in women increase the risk of transmitting mtDNA disease to their offspring.[4]



Damaging variant

A DNA sequence change that is predicted to diminish the function of the gene product (e.g., a protein or regulatory RNA).

Mitochondrial DNA (mtDNA)

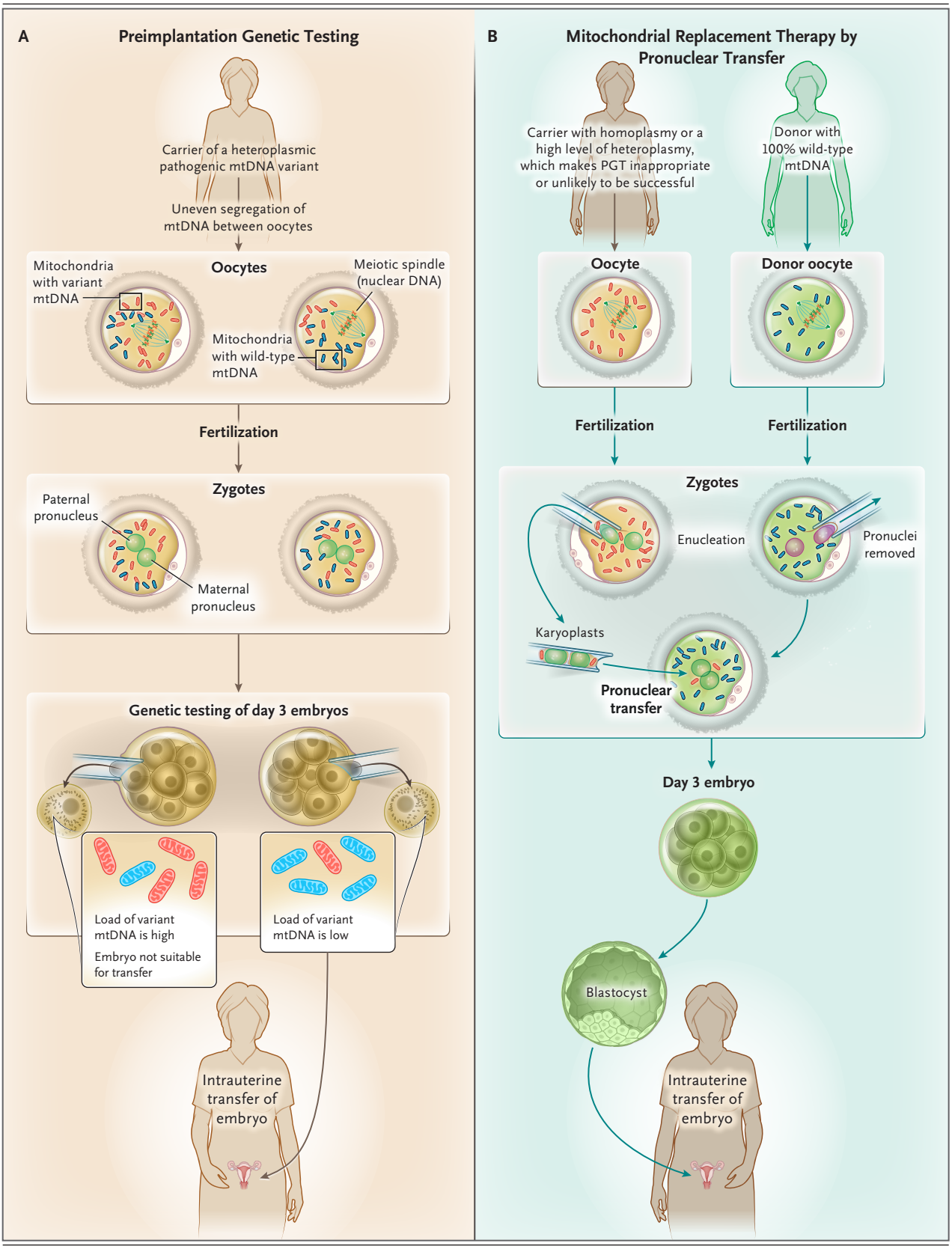
A circular DNA in the mitochondrion (there are 100 to 500,000 mtDNA copies per human cell) that encodes 22 transfer RNAs, 2 ribosomal RNAs, and 13 protein subunits, all of which are involved in the oxidative phosphorylation pathway. The mitochondrion is composed of proteins encoded by these genes, together with proteins encoded by nuclear genes. Mitochondria, and thus mtDNA and mtDNA disease, are maternally inherited. In many patients with mitochondrial disease, mtDNA with a pathogenic mtDNA variant coexists with mtDNA without the variant (a state termed heteroplasmy). Clinical symptoms depend on the extent of heteroplasmy in vulnerable tissues; symptoms occur only when the threshold for a pathogenic variant is exceeded. In some patients with mitochondrial disease, 100% of their mtDNA carries the pathogenic variant (a state termed homoplasmy)

Mitochondrion

A cytoplasmic organelle that generates, through oxidative phosphorylation, ATP, a chemical used to store and generate energy for cellular functions, including proliferation. Neurons and muscle cells are particularly sensitive to fluctuations in mitochondrial number and function, and pathogenic variants affecting mitochondrial components (such as proteins and RNAs) are often associated with encephalopathies and myopathies. The mitochondrion contains its own DNA and ribosomes with which it synthesizes in situ 13 polypeptide subunits of enzymes that mediate oxidative phosphorylation.

An illustrated glossary is available at NEJM.org

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Figure 1 (facing page). Strategies to Prevent the Transmission of Mitochondrial DNA Diseases.

Panel A shows preimplantation genetic testing, which involves quantifying the load of variant mitochondrial DNA (mtDNA) in cells biopsied from in vitro–fertilized embryos, traditionally performed on day 3 of development. Only embryos with a load below a variant-specific threshold are selected for transfer into the maternal uterus. Panel B shows mitochondrial replacement therapy involving pronuclear transfer. After fertilization, the metaphase II oocyte completes meiosis, which results in a zygote with two pronuclei — one of maternal origin and one of paternal origin. These pronuclei are removed and transferred into a donor zygote from which the original pronuclei have been removed.

PRENATAL DIAGNOSIS AND PREIMPLANTATION GENETIC TESTING

Owing to the severity of these diseases, the high risk of recurrence among siblings, and the lack of effective therapies, couples at risk for transmitting pathogenic mtDNA variants often seek prenatal diagnosis or preimplantation genetic testing (PGT). Both approaches rely on quantifying the variant mtDNA load — through analysis of amniotic fluid in prenatal diagnosis or blastomeres in PGT — to estimate the risk that the fetus or embryo will have the disease. In PGT, only embryos with a load below a variant-specific threshold are selected for transfer to the uterus. However, neither approach is a viable option for women who are homoplasmic (i.e., 100% of mtDNA carry the variant) or who have very high levels of heteroplasmy, because all their embryos are likely to harbor high variant loads.

MITOCHONDRIAL REPLACEMENT THERAPY

To overcome this limitation of prenatal diagnosis and PGT, alternative approaches such as meiotic spindle transfer and pronuclear transfer have been developed. In these techniques, the nuclear DNA from an oocyte or a zygote carrying a pathogenic mtDNA variant is removed and transferred into an enucleated donor cell that provides wild-type mtDNA (Fig. 1). Known as mitochondrial replacement therapy (MRT) or mitochondrial donation, these procedures provide hope for women with mtDNA disease who wish to have biologically related children. Nevertheless, the introduction of mitochondria from a different origin into an embryo evokes concerns.

MRT often results in embryos with heteroplasmy, owing to a small but unavoidable amount of maternal mtDNA carryover during nuclear transfer. This residual contamination poses a risk of reversion (in which the carried-over mtDNA expands and becomes a majority), as observed in certain derived embryonic stem-cell lines[5,6] and in a clinical trial conducted in Greece.[7] In this trial, MRT was investigated as a treatment for infertility in women who did not have mtDNA disease but had experienced repeated fertilization failures or early embryonic developmental arrests. This trial was based on the unproven assumption that low fertilization rates are caused by mitochondrial dysfunction in the oocyte. Of the six children born during this trial, one had high levels of maternal mtDNA. Although reversion to maternal mtDNA was not associated with adverse effects in the child, caution in interpreting this finding is warranted because heteroplasmy itself may be deleterious (especially, of course, when the mtDNA has a pathogenically variant gene). Evidence from inbred mouse models, generated by mixing mitochondria with divergent mtDNA haplotypes, has shown that, although each haplotype alone supports normal development, metabolic disorders and developmental delays may occur in heteroplasmic animals.[8,9]

Moreover, MRT combines mitochondrial and nuclear genomes from different origins, which potentially disrupts mitochondrial–nuclear crosstalk. Mice can be back-crossed to generate different lines that have the same mtDNA genome but different nuclear genomes. One such line showed a significant decrease in physical performance,[10] and another showed modified brain anatomy and impaired sensory development and behaviors,[11] probably owing to a disrupted interaction between the nuclear and mitochondrial genomes. Similar interactions have been described in disease models; for instance, mice carrying a single-nucleotide insertion in the mitochondrial gene *tRNA-Arg* have age-related hearing loss only when homozygous for the nuclear *Ahl* locus.[12] However, these findings from early studies were all obtained with the use of inbred mice, and their applicability to the genetically diverse human population remains unclear.

CAN MRT PREVENT THE TRANSMISSION OF MTDNA DISEASES?

Supporting the use of MRT in humans are the results of in vitro studies showing that MRT-

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derived human preimplantation embryos were of similar quality to unmanipulated controls.[5,6] Furthermore, gene-expression profiling showed similar patterns in MRT embryos and unmanipulated controls[5] and in derived embryonic stem cells,[13] although MRT embryos had a slightly delayed process of nuclear demethylation.[14] In 2017, the first live birth was reported after MRT by maternal meiotic spindle transfer,[15] but no long-term follow-up data have been reported for this child even though assessment of MRT safety requires rigorously controlled clinical trials with long-term monitoring of offspring, as discussed by Lovell-Badge in an editorial now appearing in the *Journal*.[16] The study described by Hyslop et al. and McFarland et al., which was rigorously conducted and involved long-term monitoring, represents a critical step toward a reconsideration of the numerous regulatory and ethical barriers against the clinical implementation of MRT in many countries, including the United States and France, where it was ruled in 2023 that MRT embryos are transgenic (i.e., with genetic modification of the germline lineage) and hence prohibited.

Indeed, the results reported by Hyslop et al. and McFarland et al., although early, support the hypothesis that MRT can prevent the transmission of mtDNA disease. Hyslop et al. reported similar percentages of women with mtDNA disease having a clinical pregnancy with MRT (36%) and with PGT (41%). Moreover, among the eight infants born after MRT, the proportion of mtDNA carrying the deleterious variant was decreased to less than 3% in all the children except two, in whom heteroplasmy levels in blood cells were 12% and 16%. These latter cases may reflect the reversion phenomenon mentioned above — but they are still well below the threshold for the development of pathologic conditions. McFarland et al. report that all eight children have developed normally.

Disclosure forms provided by the author are available with the full text of this editorial at NEJM.org.

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